

may occur, but without significantly altering the nature of these roles, and the lowest category of Not Exposed occupations, where GenAI currently has no observable potential to automate tasks.

► **Table 5. GenAI Exposure Gradients: Definition and Interpretation**

GenAI Exposure	Definition (Mean & SD of Task Scores)	Interpretation
Exposed: Gradient 4 (Highest exposure, low task variability)	$\mu \geq 0.6$ and $\mu - \sigma \geq 0.5$	High and consistent GenAI exposure across tasks within the occupation. Most current tasks in these jobs have a high potential of automation, with little variability in task-level exposure.
Exposed: Gradient 3 (Significant exposure, high task variability)	$0.5 \leq \mu < 0.6$ and $\mu + \sigma \geq 0.5$	Above-moderate occupational exposure: even though some tasks remain less exposed, the overall potential of automation of the current tasks with GenAI is growing in these occupations.
Exposed: Gradient 2 (Moderate exposure, high task variability)	$0.4 \leq \mu < 0.5$ and $\mu + \sigma \geq 0.5$	Moderate occupational AI exposure, with high task-level variability. These occupations include a mix of some tasks that are exposed to GenAI and others not at risk, making the impact uneven.
Exposed: Gradient 1 (Low exposure, high task variability)	$\mu < 0.4$ and $\mu + \sigma \geq 0.5$	Low overall GenAI exposure at the occupational level, but high variability across tasks. Some tasks within these occupations have an elevated automation potential, even if the occupation as a whole remains strongly reliant on tasks that have a low potential of automation.
Minimal Exposure (Low exposure, moderate task variability)	$\mu < 0.5$ and $\mu + \sigma > 0.4$	Occupations with low GenAI exposure, where some tasks show moderate automation potential, but overall occupational exposure remains limited.
Not Exposed	Occupations that don't meet any of the above conditions.	Occupations where most tasks remain relatively unaffected by GenAI, with low task variability and a stable low mean exposure score.

This revised framework addresses some limitations of the earlier categorization system, particularly the interpretative challenges posed by the category, “the big unknown”. By expanding the exposure categories into a continuous gradient, the updated framework improves the readability and interpretability of results and enables us to stack employment estimates into the overall category of exposed occupations (Figure 17 in Section 10).¹⁸ It also captures in a more nuanced way how GenAI can impact occupations at varying levels of exposure, based on task-level variability and overall occupational risks. Furthermore, the framework's flexibility enhances its applicability to country-specific contexts, allowing policymakers to better understand the distribution of GenAI's potential impact and prioritize interventions for the most affected groups in the national contexts.

¹⁸ One of the limitations of the current approach is that neither ISCO-08 nor the Polish classification system includes information on the relative frequency or importance of tasks within occupations. As a result, all tasks are weighted equally when calculating exposure scores. Future research could refine this by incorporating task-level weights, for instance by mapping to occupational databases such as O*NET or by collecting self-reported data on task relevance directly from workers. A similar refinement could concern mapping the level of skills to individual tasks.

We stress that these classifications merely suggest the general trends and types of occupations at different levels of exposure, since any type of task scoring system carries a degree of subjectivity. While the abilities of GenAI and its derivative technologies evolve rapidly, many additional local factors, not covered in this global study, can either speed up and expand, or limit the impact of these technologies on specific occupations. The advantage of this more detailed categorization of occupations into four progressive gradients is that it will allow for tracking the movement of occupations over time, and for occupational-level comparisons across national classification systems with varying collections of tasks.¹⁹

¹⁹ To address the weakness of missing country-level nuance in this global index, we develop an AI-based predictor described in Section 8, which captures the general scoring structure and relative score differences across 2,861 tasks in our database. We can apply this knowledge repository to predict scores for more specific tasks in national-level classification systems. For demonstrative purposes, we intend to apply this method to the context of 6-digit occupations in Poland (Gmyrek et al., forthcoming in 2025), with the possibility of replicating the method for other countries where detailed occupational descriptions are available.

► 9 Changes to occupational classifications

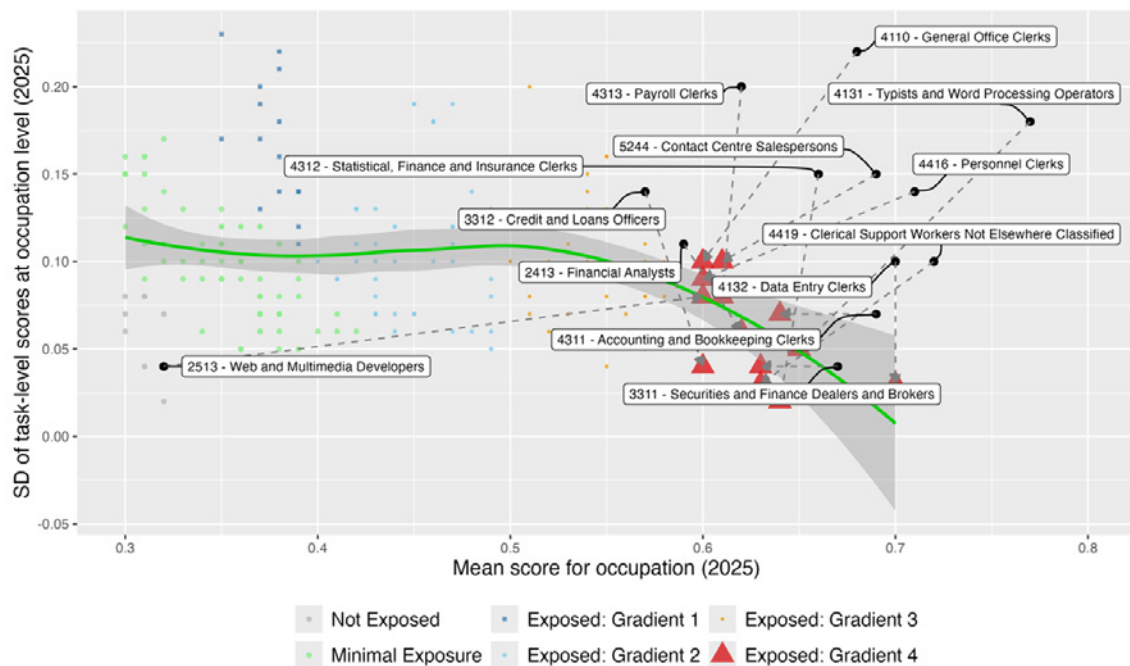
Our 2025 task-level scores are broadly consistent with those of the ILO 2023 study (Gmyrek et al., 2023), although the mean scores at the level of occupations are slightly lower (0.29 in 2025 compared to 0.30 in 2023). However, the score dispersion within occupations is considerably narrower, resulting in a more concentrated overall distribution (Figure 3).

► **Figure 17. AI Exposure: Comparison of 2025 with 2023, ISCO -08 4-digit level occupations**



Similar to our 2023 study, clerical occupations exhibit the highest levels of exposure, though with a decline in their mean scores due to a reassessment of certain task-level evaluations (Figures 18 & 19). This decrease highlights the distinction between the theoretical automation potential of a task and the practical insights gained from two years of experimentation with GenAI tools. While tasks such as taking meeting notes or scheduling appointments can significantly benefit from GenAI support, our 2023 scores for such tasks, which reached 0.9 in some cases, reflected an overly optimistic assessment of full automation potential. In reality, many of these tasks still require substantial human effort. Despite the decline in overall mean scores for these occupations, many still fall within the highest exposure gradient (Level 4), since the final classification does not depend solely on the mean score but is determined through a simultaneous assessment of each occupation's mean and the dispersion of corresponding task-level scores (SD, see Figure 16 and Table 5). Among the 13 occupations classified in gradient 4, the majority represent clerical roles, including data entry clerks, typists and word processing operators, accounting and bookkeeping clerks, and general office clerks.

► Figure 18. Changes to occupational exposure between 2023 and 2025 within Gradient 4



On the other hand, several highly digitized occupations have seen an increase in their mean scores. This applies, for example, to web and media developers, statistical and database specialists, as well as financial and software-related occupations (Figure 17). A key factor behind this shift is the rapid expansion of AI capabilities since our previous study in 2023. At that time, large language models (LLMs) were primarily viewed as advanced text generators. Since then, they have acquired multimodal abilities, allowing them to process and generate not only text but also images, audio, and video. Additionally, these models have been gaining some agentic capabilities, allowing them to execute multi-step tasks autonomously, interact with software environments, and make decisions based on contextual inputs. These advancements significantly broaden the scope of tasks that can be at least partially automated using GenAI, with software development and multimedia sectors among some of the leading adopters.

While many of these occupations fall within gradient 3 and 4, one must consider that these roles are deeply intertwined with digital technologies, and indeed, many of these occupations have emerged as a result of technological advancements. The scores reflect the high potential of automation among their current tasks, but these occupations also possess a strong capacity to evolve and develop new tasks through innovative application of GenAI and related digital tools. In previous technological transitions, this type of adaptation was exemplified by the well-known case of bank tellers following the diffusion of automatic teller machines (ATMs) in the 1990s (Bessen, 2016). As ATMs automated routine cash withdrawals, they also lowered operational costs, enabling banks to open more branches and, in turn, increasing the demand for bank tellers in roles that required more complex customer service and advisory tasks. One can therefore expect many tasks in these occupations not to remain static but rather to evolve alongside technological advancements, driven by the expanding capabilities of GenAI. A key question – largely dependent on the policies shaping this evolution – is to what extent these jobs can retrain and retain existing workers, rather than drive labor churn through workforce replacement and the recruitment of typically younger specialists (Ananian et al., 2006).